Supplementary figures - paper preview (filled to common column width, 150 dpi)

(si_performance_ranking.png)

PAIRED

FUD (↓)

Rank	CT/CS/VAS/NUT	Mean
1	● ○ ● ●	118.697
2	○ ● ● ●	121.582
3	● ○ ○ ●	129.243
...		
13	● ○ ○ ○	161.815
14	○ ● ○ ○	171.584
15	○ ○ ● ○	185.789

LPIPS (↓)

Rank	CT/CS/VAS/NUT	Mean ± SD
1	● ● ● ●	0.319 ± 0.083
2	○ ● ● ●	0.327 ± 0.083
3	● ○ ● ●	0.330 ± 0.083
...		
13	○ ○ ● ○	0.425 ± 0.068
14	○ ● ○ ○	0.429 ± 0.084
15	● ○ ○ ○	0.430 ± 0.083

PQ (↑)

Rank	CT/CS/VAS/NUT	Mean ± SD
1	● ● ○ ●	0.809 ± 0.108
2	● ● ● ●	0.808 ± 0.107
3	○ ● ● ●	0.730 ± 0.135
...		
13	○ ○ ● ●	0.133 ± 0.174
14	○ ○ ○ ●	0.131 ± 0.174
15	○ ○ ● ○	0.061 ± 0.115

DICE (↑)

Rank	CT/CS/VAS/NUT	Mean ± SD
1	● ● ○ ●	0.904 ± 0.107
2	● ● ● ●	0.903 ± 0.104
3	○ ● ● ●	0.856 ± 0.130
...		
13	○ ○ ● ●	0.277 ± 0.246
14	○ ○ ○ ●	0.267 ± 0.249
15	○ ○ ● ○	0.143 ± 0.179

HED (↓)

Rank	CT/CS/VAS/NUT	Mean ± SD
1	● ● ● ●	0.035 ± 0.023
2	○ ● ● ●	0.042 ± 0.023
3	● ○ ● ●	0.044 ± 0.028
...		
13	● ● ○ ○	0.159 ± 0.050
14	● ○ ● ○	0.160 ± 0.051
15	○ ○ ● ○	0.166 ± 0.051

UNPAIRED

FUD (↓)

Rank	CT/CS/VAS/NUT	Mean
1	○ ○ ○ ●	136.401
2	○ ○ ● ●	139.489
3	● ○ ○ ●	142.337
...		
13	● ● ○ ○	180.869
14	● ● ● ○	183.658
15	○ ○ ● ○	189.310

LPIPS (↓)

Rank	CT/CS/VAS/NUT	Mean ± SD
1	● ● ● ●	0.478 ± 0.121
2	○ ● ● ●	0.491 ± 0.119
3	● ○ ● ●	0.492 ± 0.120
...		
13	● ○ ○ ○	0.547 ± 0.113
14	○ ○ ○ ●	0.549 ± 0.102
15	○ ○ ● ○	0.564 ± 0.104

PQ (↑)

Rank	CT/CS/VAS/NUT	Mean ± SD
1	● ● ○ ●	0.765 ± 0.167
2	● ● ● ●	0.757 ± 0.164
3	● ● ○ ○	0.681 ± 0.159
...		
13	○ ○ ○ ●	0.140 ± 0.115
14	○ ○ ● ●	0.134 ± 0.110
15	○ ○ ● ○	0.064 ± 0.083

DICE (↑)

Rank	CT/CS/VAS/NUT	Mean ± SD
1	● ● ○ ●	0.847 ± 0.193
2	● ● ● ●	0.839 ± 0.195
3	● ● ○ ○	0.827 ± 0.153
...		
13	○ ○ ○ ●	0.308 ± 0.175
14	○ ○ ● ●	0.298 ± 0.171
15	○ ○ ● ○	0.171 ± 0.153

HED (↓)

Rank	CT/CS/VAS/NUT	Mean ± SD
1	● ● ● ●	0.044 ± 0.031
2	○ ● ● ●	0.046 ± 0.026
3	● ○ ● ●	0.049 ± 0.030
...		
13	● ● ● ○	0.160 ± 0.053
14	● ○ ● ○	0.162 ± 0.055
15	○ ○ ● ○	0.172 ± 0.054

Figure S1. Top- and bottom-ranked channel-group conditions per metric, for paired and unpaired inference. Top-3 and bottom-3 ranked conditions per metric — FUD (↓), LPIPS (↓), PQ (↑), DICE (↑), HED (↓) — for paired (upper) and unpaired (lower) inference. The dot column encodes active channel groups in the order CT, CS, VA, NU, UNI (filled = included): CT (Cell types) groups the healthy / cancer / immune density maps; CS (Cell state) groups the proliferative / nonproliferative / dead density maps; VA (Vasculature) is the vasculature density channel; NU (Nutrient) groups the oxygen and glucose channels; UNI is the UNI-2h H&E reference embedding (paired-only). The mean and per-tile mean ± SD are listed per ranked condition.

(ablation_grids_combined.png)

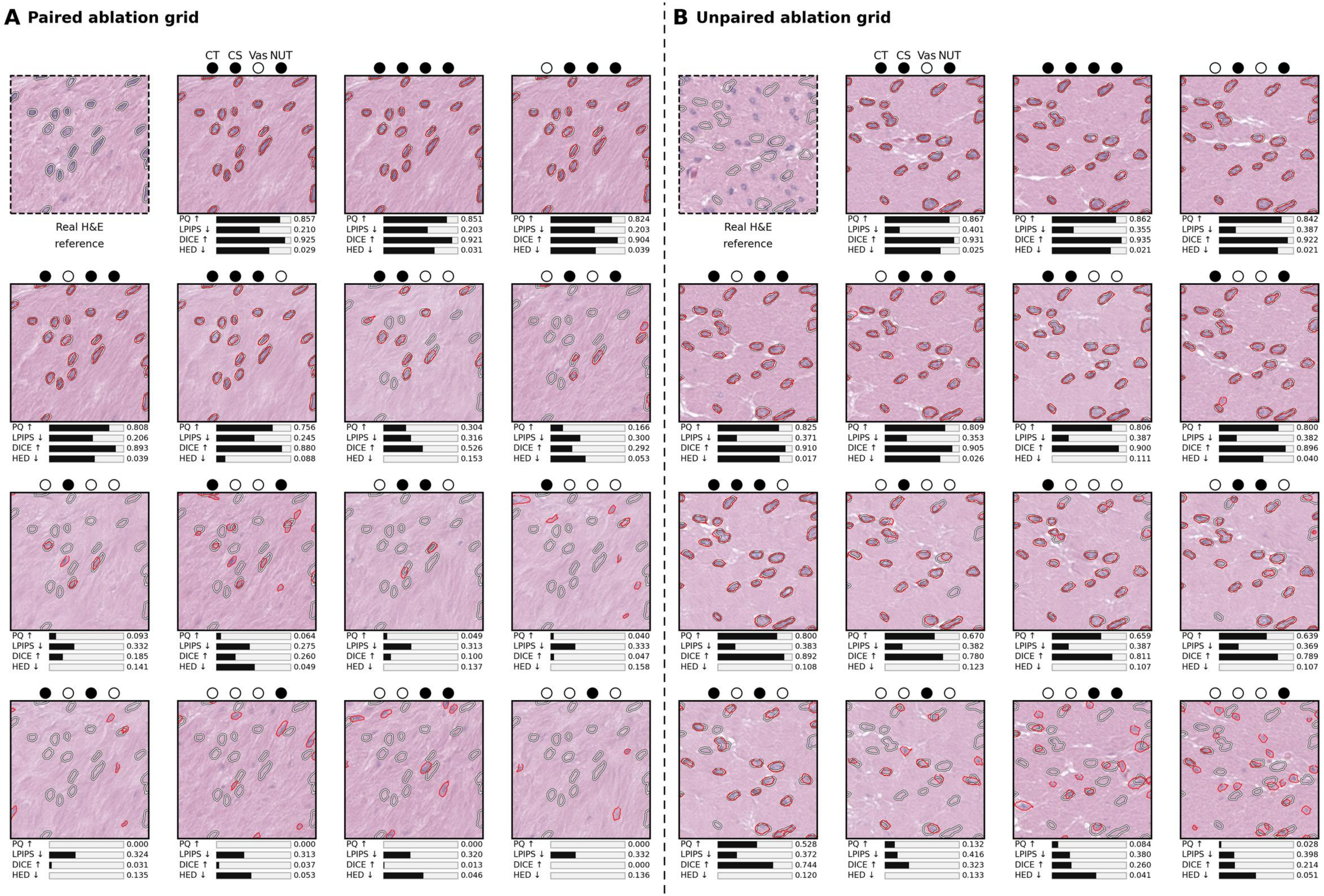

Figure S2. Generated H&E patches under channel-group ablations illustrate ControlNet output for paired and unpaired inference. (A) Paired-inference results for one representative tile. **(B)** Unpaired-inference results for one representative tile. Each block is a 4 × 4 grid; the first cell (gray dashed border, top-left) is the real H&E reference for that tile, and the remaining 15 cells are generated patches sorted by Panoptic Quality (PQ; higher is better) from best to worst across channel-group on/off combinations. The 4-dot header above each cell encodes active channel groups in the order CT, CS, VA, NU (Cell types, Cell state, Vasculature, Nutrient); a filled dot indicates the group is included. Per-cell bar plots report LPIPS (lower is better), PQ (higher is better), DICE (higher is better), and HED (lower is better) for that single patch, with the numeric value printed beside each bar.

(si_a1a2_qualitative_tiles.png)

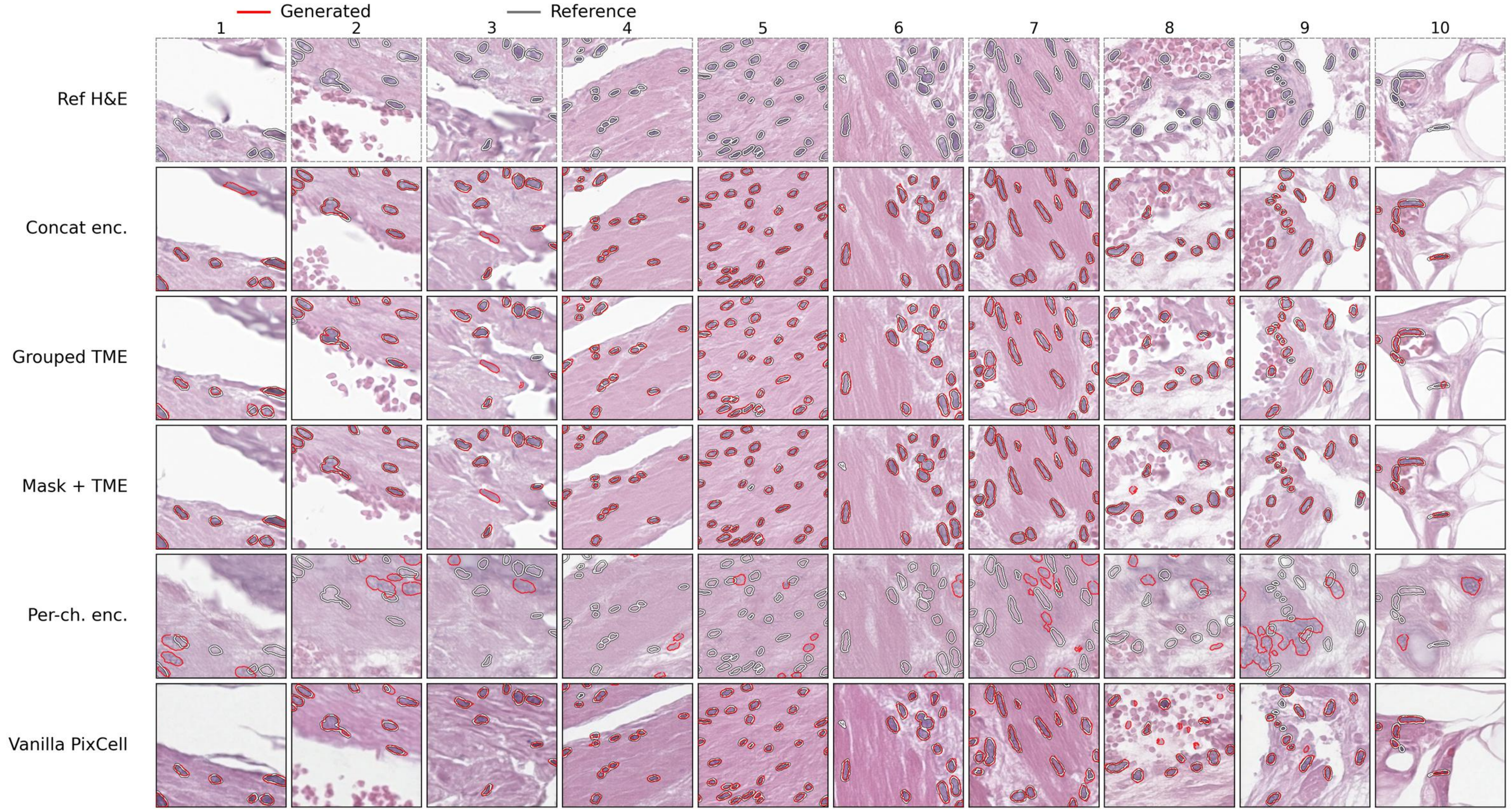

Figure S3. Conditioning-architecture ablation: qualitative H&E for each TME-injection variant. Qualitative tiles across 10 columns for the conditioning-architecture variants — Grouped TME only, Concat TME encoder, Per-channel TME encoders, additive Mask + TME, and an off-the-shelf Vanilla PixCell ControlNet baseline. The top row is the real reference H&E; each lower row is one variant's generated H&E, with overlaid contours marking segmented nuclei (red = generated, gray = reference) so segmentation fidelity can be compared per tile. The full per-variant metrics table (FUD ↓, DICE ↑, PQ ↑, LPIPS ↓, HED ↓ and trainable parameter count) is provided separately as `individual/si_a1_a2/SI_A1_A2_section2_metrics.png`.